

Computing Methods for Physics – 5 July 2022

Your exam material (code files, plots, datafiles, etc.) must be submitted via google classroom by 13:30 as a single zip file.

C++ evaluation will be based on: correct syntax, proper return types, proper arguments of functions, data members and class interfaces, comments throughout the code, separation of class implementations and interfaces.

Python evaluation will be based on: correct syntax, avoiding C-style loops, using Python features in general, comments throughout the notebook/scripts, labels, legends and plot styling and clarity in general.

Part 1 – Classes, inheritance, and polymorphism in C++

Implement a C++ class `DataSet` to represent and manipulate N positive, real numbers. As private data members, your class must have at least N and the values $\{x_i > 0, i = 1, \dots, N\}$. The class must also provide the following.

- A constructor that takes as arguments N and the path to a text file that stores N values $\{x_i\}$ (this constructor must read in the data stored in the file).
- A copy constructor.
- A method to add a new data point to an existing `DataSet` instance, and a method to remove a data point from an existing `DataSet` instance.
- An overload of the plus (+) operator to shift all x_i 's by a given quantity (e.g., if the instance `D1` of `DataSet` is storing $\{1, 5, 1, 2\}$, then `D1 + 3` must be an instance of `DataSet` that stores $\{4, 8, 4, 5\}$).

Finally, provide three classes `Arithmetic`, `Geometric` and `Harmonic` that have `DataSet` as parent class and implement the method `mean()`. As a reminder these means are defined as

$$A = \frac{1}{N} \sum_{i=1}^N x_i$$
$$G = \left(\prod_{i=1}^N x_i \right)^{1/N}$$
$$H = \frac{N}{\sum_{i=1}^N x_i^{-1}}.$$

Your submitted material must include a file `app.cpp` that showcases the classes you implemented.

Part 2 – Classes, inheritance, and polymorphism in Python

Use a Python notebook or Python scripts to complete the following tasks. Aim for clean, well commented and efficient code.

1. Draw $N = 10^4$ random values from a Gaussian distribution with centre μ and width σ of your choice. Store these values to a text file, so that you can use them to test your C++ code.
2. Design a class to process an array of integers in order to find pairs of its elements (identified by their indices) such that their sum is equal to a specific number s . [The array and s are arguments of the class constructor.] Test your class with 10^2 randomly drawn integers.
3. Design a class to find triplets of elements in an array of random integers such that their sum is zero. [The array is an argument of the class constructor, and the triplets are identified by their indices within the array.] Test your class with 10^3 random integers.
4. Design a class that inherits from the two classes described above and is able to handle either problem, given the user's request that must be provided interactively [i.e., ask the user a question and process the answer appropriately].